Data Governance Maturity Assessment: A Case Study in IT Bureau of Audit Board

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***Abstract*—Data are important assets of organizations and define their businesses or functions. Data needs to be governed and controlled so that it goes to the proper owner. Well-defined data management can help organizations maintain the quality of their data. This study took the IT Bureau of Audit Board of State as a study research. The study assesses its maturity level and recommend how the board can benefit from learning it. This study used Stanford Maturity Model as the framework tool to assess practices of data governance in IT Bureau of audit board. The maturity assessment is performed by assessing both IT Bureau staffs and top echelons. After validation and finalization assessment, the result showed that most maturity components had average maturity level of 2.63. It means the IT Bureau of Audit Board already has high-level data governance meta-policies which are considered as best practices (between level 2 and 3). The data governance components that can be improved in the future are stewardship and metadata on capabilities dimension and policies dimension.**

Keywords—data management, stanford maturity model, local government, data governance

I. INTRODUCTION

Data is often being overlooked because it is considered difficult to manage. Organizations have the potential to adapt better if they know the significance of managing data and recognized them as tangible asset [1]. A collection of data form information and knowledge and they can be useful or misleading. The quality of those data depends on whether the data is well managed or not. Thus, data need to be planned, managed, and controlled through data management (DAMA).

There are a lot of functions of DAMA based on DMBOK, and the core of those functions is data governance (DG). Data needs to be governed and controlled so that it goes to the proper owner. Data is important but also need to be secured. A study shows that being ignorant about security in data and improper DG practices can harm business organization financially [2]. Study about Local government in Malaysia confirmed that there are two main crucial factors for acceptance of its master data management: DG and data quality [3]. Further study suggested that DG is also needed outside of organizations. Recently study about digital governance is being researched. Digital governance is a set of actions about enacting procedures and policies for the use of digital data globally [4]. By having well defined DG and good implementation, any organizations or entities should have secured and decent quality data and information as the output.

This study observes the IT (Information Technology) Bureau within Audit Board of State as research subject. The IT Bureau of Audit Board has several tasks and functions, which are executive data steward, database administrator, database consumer, and database monitor. Audit Board works closely with data and its IT Bureau is responsible managing it. As Audit Bureau, knowing how important DG as stated previously is, they need to know how mature their DG implementation so that they can improve it accordingly. Getting to know how mature its DG, the Audit Board can adjust and reorganize their strategy if they hope to reach some maturity level of DAMA in DG based on the assessment. An organization need maturity model framework as best practice method that can be used to assess the baseline. The maturity model usually consists of some methods and components which need to be used accordingly.

This study is organized in the following structure. The first section introduces background of the study. In the second part, the related studies will be reviewed. The assessment steps will be explained in the third section. Lastly, the result of study will be analyzed and discussed in conclusion.

II. LITERATURE STUDY

Data management science has a very broad scope. there are so many references related to it, but this study limits only the data governance, data governance maturity model and more specifically the Stanford University DG maturity model.

A. Data Governance

Data governance implement how to plan, monitor, and enforce (as overseeing and authority activities) toward management of data and information assets. It guides implementation of all data management functions [1]. DG is the center of data management framework which ensure data is precise, shared properly, and secured [6].

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Fig. 1. Data Governance on Data Management Framework [1]

As shown in Figure 1, DAMA summarizes 10 main data management functions. They include managing data architecture, managing data modeling & design, managing data storage & operations, managing data security, managing data integration & interoperability, managing document & content, managing reference and master data, managing data warehouse & business intelligence, managing metadata, and managing data quality [1].

B. Data Governance Maturity Model

Maturity assessment have crucial role to oversee, assess and upgrade capacity of data governance within an organization. There are two methods to perform it: self-evaluation and third-party (external) independent evaluation. Organizational self-evaluation is performed to implement self-accomplishment, self-examination, and sustainable improvement. It also arrange employee (on internal circle) as the center of DG evaluation activities and system [6].

Whilst, a DG model of maturity is a technic used to build, improve and evaluate an extensive program. It is a significant technic for designing and maintaining the next strategy of program [7]. A crucial benefit of using a maturity model is that it can assess consistently the development progress of a program [8].

Most maturity model definitions found in literature clarify that they are key factor in knowing potencies (weaknesses and strengths) of the context of organization to what they are adopted, and the information by methodologies which connected with benchmarking [10]. Each maturity model has different emphasis, scope and focus depending on the purpose of maturity assessment as shown in Table I.

TABLE I. MATURITY MODEL EMPHASIS

Organization	Emphasis
CMMI [15]	Process characterized for organizational applicability
IBM [13][14]	Process characterized for organizational applicability
Data Flux [16]	Organization, Risk, Reward, Process, Technology, Adoption, and Business Capabilities
Kalido [13]	Perceived data value, Risk, Reward, Data as enterprise asset
Stanford [7]	Focuses in 3 dimensions: People, Policies, Capabilities with components includes Awareness, Metadata, Formalization, Data Quality, Stewardship, and Master Data

With the purpose of identifying fundamental aspects in implementing DG, these are 4 common domains identified from a selected maturity models:

1. Data Principles: Systematized policy on collecting, keeping, and removing data [14], assisted by analytic and sustainable calculation [15]. Defining "acceptable" data using business policies guiding collection and management of data [16] that are formally defined, implemented and enforced within the organization [6].
2. Data Lifecycle: Systematized policy on collecting, keeping, and removing data [14], assisted by analytic and sustainable calculation [15].
3. Data Quality: calculation enhancement and quality certification data [14] avoiding loss to operational and reputational levels [15]. A continuous process to define criterion for specifying acceptable levels of data quality corresponding business requirements, and assure data quality reaches these levels [6].
4. Metadata: describing what data is about and providing a common, single and consistent semantic explanation. It also establishes a competence at technical and business level [14], as well as their use, representation, context and interrelationship that explains what, who, when, where, why and how to data consumers and more IT properties[6].

TABLE II. MATURITY MODEL DOMAIN COMPARISON

Organization	Domains			
	Data Principles	Data Lifecycle	Data Quality	Metadata
CMMI [15]	✓		✓	✓
IBM [13][14]	✓	✓	✓	✓
Data Flux [16]	✓			
Kalido [13]	✓	✓		
Stanford [7]	✓		✓	✓

Based on the comparison in Table II, Stanford Maturity Model has comprehensive approach in evaluating the main domains in DG focusing in people, policies and capabilities aspects to determine the DG maturity level of the organization.

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A previous study by Permana and Suroso (2018) has used Stanford Maturity Model to assess the DG maturity level of data management division of a company. The research aims to recommend a solution on data governance problems based on the result of the assessment conducted [10].

This study chose Stanford Maturity Model because it was considered capable to describe circumstance of the data governance maturity level from the research subject.

C. *Stanford University's Data Governance Maturity Model (Stanford DGMM)*

The Stanford DGMM was designed by Stanford University with the specific targets, competencies, and preferences of the institution. Many aspects of this DG model were adapted from other DG maturity models and customized for the program being developed [8].

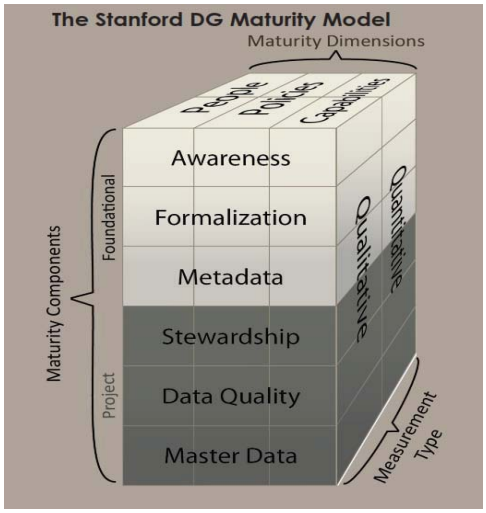


Fig. 2. Stanford DG Maturity Model [7]

The maturity model consists of 2 core components (on Fig.2): Foundational and Project Components. Competencies of core DG and critical program resources development can be measured by the foundational components with 3 subjects [6][7]:

- Awareness: The individual comprehension and sense within the organization knowledge of the regulations, roles, and technologies associated with the DG program.
- Formalization: how structured roles in an organization and the activities of the employees are governed by regulations.
- Metadata: Describing other data and IT assets by relating essential business and technical information. Also describing data which facilitates the consistent understanding of the characteristics and usage of data. Business metadata answers who, what, where, when, why and how for users of the data and other IT properties or assets.

The Project Component on the other hand focuses on measuring effectiveness of DG concepts applied in the course of sponsored projects with 3 subjects [6][7]:

- Stewardship: Formalization of accountability for the description, utilization, and standard of quality of specific data assets within a defined organizational context.
- Data Quality: sustainable process for defining the criterion on specifying acceptable levels of data quality to reach business needs, and on ensuring that data quality reach these levels.
- Master Data: Business-critical data that is highly shared across the organization. Master data are often codified data which describe the structure of the organization or key data entities.

There are 3 dimensions that split each of the six components above (awareness, formalization, metadata, stewardship, data quality, and master data), focusing on particular aspects of component development [7].

- People: Roles and organization structures.
- Policies: Development, auditing and enforcement of data policies, standards and best practices.
- Capabilities: Enabling technologies and techniques.

To assess the maturity level, each component will be evaluated using the DG Maturity Tool. The Tool is used to determine the level each component (Foundational and Project) which is also subdivided into 3 dimensions mentioned before (People. Policies and Capabilities). Each dimensions consists of 5 maturity level stages. These level will be mapped into a scorecard then to calculate the overall that will reveal maturity level based on dimensions and component. The level of organization maturity in each area [6].

An initial assessment of organization should be executed early in the DG step of program, ideally during the planning phase. On this phase the frequency of assessment is also determined. Its Frequency relies on many factors, such as the available resources to perform DG program, or the maturity of DG program on current time. Mostly the recommendation suggest DG program be assessed at least every year to find out the maturity progress of the organization's DG implementation [6].

III. Methodology

This research conduct 4 steps which are interview and initial assessment, data compiling and analyzing, confirmation interview and finalizing assessment and then finishing the report.

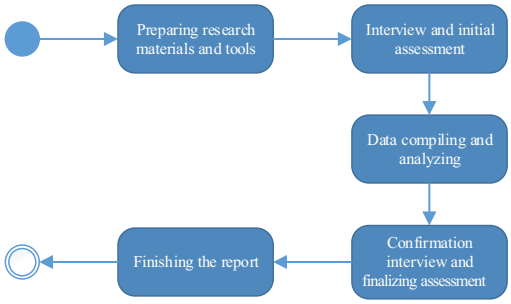


Fig. 3. Steps of methodology

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The steps by step of the research are as follow:

A. Preparing research materials and tools

Before this assessment was carried out, materials and tools need to be prepared to assist in getting the data needed. The research material was obtained from Stanford Data Governance Maturity Model, afterward adjusted to the assessment needs.

B. Interview and initial assessment

On this step, this study performed interview to IT Bureau management to identify the scope of organization. It determined the method to be used. After selecting the method, this study conducted initial assessment by asking for opinions on the data governance key people. The opinions were collected through filling data governance maturity questionnaire. Of 50 bureau staffs in all divisions, the questionnaire was distributed to 20 key people that were database administrator, database consumer, and database monitor.

C. Data compiling and analyzing

After getting the results from the previous step, the data was compiled and analyzed. The result would be summarized, averaged and sorted. This result can be called as initial assessment result.

D. Confirmation interview and finalizing assessment

The compiled and analyzed result would be confirmed to IT Bureau management by interviewing IT Bureau leader. This interview would also directly affect the previous result of the assessment because the IT Bureau leaders are considered capable to evaluate the circumstance on broader scope.

E. Finishing the report

This step finalized all analysis and produced the result as the confirmed assessment result.

IV. Result and Discussion

The planned steps were executed sequentially. The scoping interview was held to determine the method to be chosen, and the initial assessment followed as matter to confirm to IT Bureau management. Detail of these steps are explained as follows.

A. Conducting interview and first assessment

Interview and questionnaire was conducted on key people that work as IT Bureau management, database administrator, database consumer and database monitor. The interview was conducted with management, and the questionnaire was distributed to the bureau staff that has role as database administrator, database consumer or database monitor. They carry out their roles in accordance with the prevailing job procedures [11]. There were 20 questionnaires that had been distributed, 8 responses were obtained.

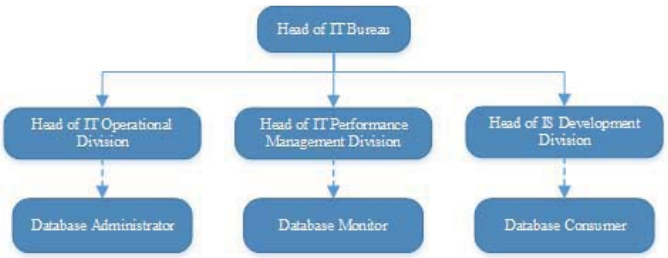


Fig. 4. Structures of data governance on IT Bureau

The interview was conducted to discuss several areas, one of which was the complexity of data management in the IT Bureau. Based on that point, this study concluded that the IT Bureau has a simple data management complexity (see Fig 4). So, this research needs a simple method but comprehensive to assess the data governance maturity of IT Bureau, and Stanford Data Governance Maturity Model is chosen [12].

B. Compiling and analyzing the initial assessment result

The questionnaire was distributed to 8 IT Bureau data governance key people. The result of initial assessment is drawn on TABLE III.

TABLE III. Result of Initial Assessment

No	Component	Level				
		1	*2*	*3*	*4*	*5*
Foundational Component						
	Awareness					
1	People		4	3		1
2	Policies	2	2	4		
3	Capabilities		4	2		2
	Formalization					
4	People		3	3	2	
5	Policies	2	3	3		
6	Capabilities	2	2	4		
	Metadata					
7	People		4	1	3	
8	Policies	4		4		
9	Capabilities	3	3	2		
Project Component						
	Stewardship					
10	People	3	3	2		
11	Policies	4	2	2		
12	Capabilities	4	2	1	1	
	Data Quality					
13	People	4	2	2		
14	Policies	3	1	3	1	
15	Capabilities	2	3	3		
	Master Data					
16	People	1	3	3	1	
17	Policies	7		1		
18	Capabilities	2	2	2	2	

The raw results of initial assessment are averaged to simplify reader to see illustration of data governance maturity model.

TABLE IV. Mean of Initial Assessment

Foundational Component	People	Policies	Capabilities	Mean
Awareness	2,75	2,25	3,00	2,67
Formalization	2,88	2,13	2,25	2,42

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Metadata	2,88	2,00	1,88	2,25
Mean	2,83	2,13	2,38	**2,44**

Project Component	People	Policies	Capabilities	Mean
Stewardship	1,88	1,75	1,88	1,83
Data Quality	1,75	2,25	2,13	2,04
Master Data	2,50	1,25	2,50	2,08
Mean	2,04	1,75	2,17	**1,99**

The Result of TABLE IV represents opinion of IT Bureau data management key people about data governance maturity.

There are several information that can be obtained from this initial assessment. As seen in table IV, the lowest maturity level is master data component on policies dimension with value 1.25 and followed respectively by stewardship component on policies dimension and data quality component on people dimension both with value 1.75. The highest maturity level is awareness component on capabilities dimension with value 3.00 and followed respectively by formalization component and metadata component both on people dimension with value 2.88. This illustration is in line with the results of the averaged dimension of each component which obtain the lowest maturity level for component is stewardship component with value 1.83 and the highest maturity level for component is awareness component with value 2.67.

The foundational component mean and project component mean (the number that is bold) are combined and averaged, they obtained data governance maturity level on opinion of IT Bureau data management key people with value 2.22.

C. Confirmation interview and Finalizing assessment

By using the initial assessment as base reference, this study continues to confirmation interview. The interviews were conducted toward each division leader (3 divisions). They were asked to decide the level of maturity on each question of data governance maturity assessment by considering the result from initial assessment.

The result value of initial assessment which conduct to data management key people was only used as a reference and is used as a comparison only. The confirmation result value is calculated based on confirmation interview (Step D in section III). The summary result is shown on TABLE V.

TABLE V. RESULT OF CONFIRMATION ASSESSMENT

No	Component	Level				
		1	*2*	*3*	*4*	*5*
Foundational Component						
	Awareness					
1	People		1	1	1	
2	Policies		2	1		
3	Capabilities		1	1	1	
	Formalization					
4	People		1	1	1	
5	Policies	1	1		1	
6	Capabilities		1	1	1	
	Metadata					
7	People		1	1	1	
8	Policies	1	1	1		
9	Capabilities	1		2		
Project Component						
	Stewardship					
10	People		2		1	
11	Policies	1	1	1		
12	Capabilities	1		2		
	Data Quality					
13	People		1	1	1	
14	Policies		1		2	
15	Capabilities		2	1		
	Master Data					
16	People		2	1		
17	Policies	1	1		1	
18	Capabilities			1	2	

The average of confirmation result is outlined on TABLE VI.

TABLE VI. AVERAGED CONFIRMATION ASSESSMENT

Foundational Component	People	Policies	Capabilities	Mean
Awareness	3,00	2,33	3,00	2,78
Formalization	3,00	2,33	3,00	2,78
Metadata	3,00	2,00	2,33	2,44
Mean	3,00	2,22	2,78	**2,67**

Project Component	People	Policies	Capabilities	Mean
Stewardship	2,67	2,00	1,67	2,11
Data Quality	3,00	3,33	2,33	2,89
Master Data	2,33	2,33	3,67	2,78
Mean	2,67	2,55	2,56	**2,59**

The information that can be obtained from this confirmation result are:

1) The lowest maturity level is stewardship component on capabilities dimension with value 1.67.
2) The highest maturity level is master data component on capabilities dimension with value 3.67.
3) The lowest maturity level is on stewardship component with the averaged value of dimensions is 2.11.
4) The highest maturity level is on data quality component with the averaged value of dimensions is 2.89.
5) The overall data governance maturity level based on confirmation interview on IT Bureau Leaders is 2.63. This value is the mean value of foundational and project components value.

The result of TABLE IV and TABLE VI can be pictured by following radar chart:

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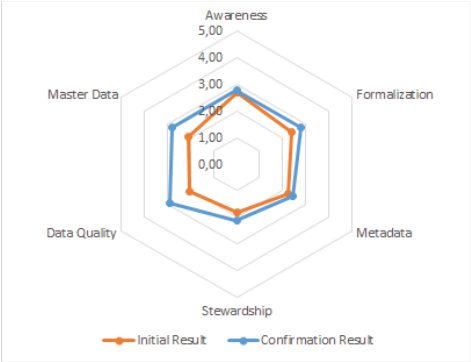


Fig. 5. Radar chart of comparison of initial and confirmation result

D. Discussion

Based on the result the division leaders give higher scores almost on all component's dimensions. Except for the awareness component on capabilities dimension and metadata component on policies dimension that have same score, and stewardship component on capabilities dimension and master data component on people dimension with lower scores but just with bit differences (1,67 with 1,88 and 2,33 with 2,50).

There are several component's dimension that have significant gap with staffs result with more than 1 point gap. The data quality component on people dimension has 1,25 point gap (3,00 with 1,75) and on policies dimension has 1,08 point gap (3,33 with 2,25). The master data component on policies dimension has 1,08 point gap (2,33 with 1,25) and on capabilities dimension has 1,17 point gap (3,67 with 2,50). Interesting that these two component are part of project component. Need further studies to analyze the cause of these significant gap.

This study also founded that the stewardship is still weak, and significant improvements are needed to increase the maturity level on this sector. One of the improvement can be held by mapping the stewardship dimensions, which means that operational and coordinative side must be clearly defined. Stewarding data and organizing governance have significant function on setting procedures, standards, policies, and architecture, and they also play a role on resolving data related issues [1]. Another improvement is by increasing the control of stewardship processes. This means that there must be an initiator who monitors the stewardship process. Well running of stewardship process is one of the keys to success of good data governance [1]. And then the improvement can be fixing the data governance key people's perspective that stewardship is a full time role. It means that stewardship is considered as a part time role, otherwise the goal of the stewardship will not be achieved.

V. Conclusion

From the result obtained, this study concludes that the data governance maturity level of the IT Bureau of Audit Board according to the Stanford Data Governance Maturity Assessment method is at the level 2.63. This level value is obtained from the result based on confirmation interview on IT Bureau leader. Even though the IT Bureau leader is considered able to see the circumstance more broadly, the results of the initial assessment and confirmation assessment are not much different, still within the level range between 2 and 3.

Although this study concludes that data quality is the strongest component of IT bureau of audit board but it was one of the component that have significant gap with the initial assessment's result. The improvements are must be carried out consistently and continuously. There are no instant changes. data quality management need to be continuously improved, because data quality is the goal of the entire data management process.

References

[1] DAMA (2017). DAMA-DMBOK : Data Management Body of Knowledge (2nd ed.). Bradley Beach: Technics Publications.

[2] S. Kamariah Abu, A. Norshakirah, R.Ade Wahyu, and H. Noor Hafzah, "Data Governance Cloud Security Assessment at Data Center", 4th International Conference on Computer and Information Sciences (ICCOINS), 2018.

[3] H. Faizura, K. Nazri, T. Nazim, P. David, and Nur Azaliah Abu Bakara,"Determinants of master data management adoption by local government organizations: An empirical study" in International Journal of Information Management, 2019.

[4] F. Luciano, "Soft ethics, the governance of the digital and the General Data Protection Regulation", Phil. Trans. R, 2018.

[5] Cheng G, et al,"Cloud Data Governance Maturity Model" in 8th IEEE International Conference on Software Engineering and Service Science (ICSESS), 2017.

[6] OMES (Office of Management and Enterprise Services),"Data Governance Maturity Model", 2011.

[7] Stanford University ,"Data Governance at Stanford: The Stanford Data Governance Maturity Model", 2011, [Online] "https://web.stanford.edu/dept/pres-provost/cgi-bin/dg/wordpress/wp-content/uploads/2011/11/DG-News002.pdf"

[8] A. Fitrian, M. Andri Akbar, and A. Nur, "Forest and Land Use Governance in a decentralized Indonesia, a legal and policy review"

[9] D. Proença, J. Borbinha,"Maturity Models for Data and Information Management", 2018.

[10] R.I. Permana, J.S. Suroso, "Data Governance Maturity Assessment at PT. XYZ. Case Study : Data Management Division", International Conference on Information Management and Technology (ICIMTech), 2018.

[11] Supreme Audit Board of Indonesia, "Organizational Structure and Job Procedures", 2014.

[12] Stanford University, "Data Governance Maturity Model Guiding Questions for each Component-Dimension", 2011, [Online] "http://web.stanford.edu/dept/pres-provost/cgi-bin/dg/wordpress/wp-content/uploads/2011/11/StanfordDataGovernanceMaturityModel.pdf"

[13] Rivera S, Loarte1 N, Raymundo C, Dominguez F, "Data Governance Maturity Model for Micro Financial Organizations in Peru", 2017, Proceedings of the 19th International Conference on Enterprise Information System (ICEIS).

[14] IBM, "Data Governance Council Maturity Model:"Building a roadmap for effective data governance",2007, [Online]. Available at: https://www-935.ibm.com/services/uk/cio/pdf/leverage_wp_data_gov_council_maturity_model.pdf

[15] CMMI Institute, "Data Governance Maturity Model", 2014, .[Online]. http://www.fstech.co.uk/fst/whitepapers/The_Data_Governance_Maturity_Model.pdf

[16] DataFlux Company, "Data Governance Maturity Model", 2010, [Online],"http://www.fstech.co.uk/fst/whitepapers/The_Data_Governance_Maturity_Model.pdf.

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